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Day_10_Task_5.ipynb | Day11_Task_2.ipynb | sample_student_performanc

Code | Python 3 (ipykernel)

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("ecommerce_dirty_sample.csv")
```

[2]:
```
df.head()
```

[2]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city
0	ORD1001	2026-01-03	CUST001	Laptop	Electronics	1	899.99	899.99	Credit Card	Shipped	Chennai
1	ORD1002	2026-01-04	CUST002	wireless mouse	Electronics	2	19.99	39.98	UPI	Delivered	Bengaluru
2	ORD1003	2026-01-05	CUST003	Office Chair	Furniture	1	149.50	149.50	Cash on Delivery	Delivered	Mumbai
3	ORD1004	2026-01-06	CUST004	Smartphone	Electronics	1	699.00	699.00	NaN	Processing	Hyderabad
4	ORD1005	2026-01-07	CUST005	Coffee Maker	Home Appliances	2	45.00	90.00	UPI	Delivered	Pune

[3]:
```
df.shape
```

[3]:
```
(104, 11)
```

[4]:
```
df.columns
```

[4]:
```
Index(['order_id', 'order_date', 'customer_id', 'product_name', 'category',
       'quantity', 'unit_price', 'total_amount', 'payment_method',
       'order_status', 'city'],
      dtype='str')
```

[5]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 104 entries, 0 to 103
Data columns (total 11 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   order_id        104 non-null    str
 1   order_date      103 non-null    str
 2   customer_id     101 non-null    str
 3   product_name    103 non-null    str
 4   category        103 non-null    str
 5   quantity        104 non-null    str
 6   unit_price      102 non-null    float64
 7   total_amount    102 non-null    float64
 8   payment_method  103 non-null    str
 9   order_status    103 non-null    str
 10  city            103 non-null    str
dtypes: float64(2), str(9)
memory usage: 9.1 KB
```

[6]:
```
df.isnull().sum()
```

[6]:
```
order_id          0
order_date        1
customer_id       3
product_name      1
category          1
quantity          0
unit_price        2
total_amount      2
payment_method    1
order_status      1
city              1
dtype: int64
```

[7]:
```
df.describe()
```

[7]:

	unit_price	total_amount
count	102.000000	1.020000e+02
mean	9976.214706	9.824241e+04
std	98997.900387	9.901273e+05
min	-45.000000	0.000000e+00
25%	22.500000	4.800000e+01
50%	59.990000	8.997000e+01
75%	149.500000	1.495000e+02
max	999999.990000	1.000000e+07

[8]:
```
df.duplicated().sum()
```

[8]:
```
np.int64(3)
```

[9]:
```
df[df.duplicated()]
```

[9]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city
100	ORD1002	2026-01-04	CUST002	wireless mouse	Electronics	2	19.99	39.98	UPI	Delivered	Bengaluru
101	ORD1009	not-a-date	CUST009	Desk Lamp	Home Decor	2	24.50	49.00	Debit Card	Delivered	Jaipur
102	ORD1016	2026-01-18	CUST016	laptop	Electronics	1	899.99	899.99	credit card	Shipped	Chennai

[10]:
```
df = df.drop_duplicates()
```

[11]:
```
df[df.duplicated()]
```

[11]:

order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city

[12]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 101 entries, 0 to 103
Data columns (total 11 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   order_id        101 non-null    str
 1   order_date      100 non-null    str
 2   customer_id     98 non-null     str
 3   product_name    100 non-null    str
 4   category        100 non-null    str
 5   quantity        101 non-null    str
 6   unit_price      99 non-null     float64
 7   total_amount    99 non-null     float64
 8   payment_method  100 non-null    str
 9   order_status    100 non-null    str
 10  city            100 non-null    str
dtypes: float64(2), str(9)
memory usage: 9.5 KB
```

[13]:
```
df["order_date"] = pd.to_datetime(df["order_date"], errors="coerce")
df["quantity"] = pd.to_numeric(df["quantity"], errors="coerce")
df["unit_price"] = pd.to_numeric(df["unit_price"], errors="coerce")
df["total_amount"] = pd.to_numeric(df["total_amount"], errors="coerce")
```

[14]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 101 entries, 0 to 103
Data columns (total 11 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   order_id        101 non-null    str
 1   order_date      97 non-null     datetime64[us]
 2   customer_id     98 non-null     str
 3   product_name    100 non-null    str
 4   category        100 non-null    str
 5   quantity        99 non-null     float64
 6   unit_price      99 non-null     float64
 7   total_amount    99 non-null     float64
 8   payment_method  100 non-null    str
 9   order_status    100 non-null    str
 10  city            100 non-null    str
dtypes: datetime64[us](1), float64(3), str(7)
memory usage: 9.5 KB
```

[15]:
```
df.isnull().sum()
```

[15]:
```
order_id          0
order_date        4
customer_id       3
product_name      1
category          1
quantity          2
unit_price        2
total_amount      2
payment_method    1
order_status      1
city              1
dtype: int64
```

[16]:
```
df["product_name"] = df["product_name"].str.strip().str.title()
df["category"] = df["category"].str.strip().str.title()
df["payment_method"] = df["payment_method"].str.strip().str.title()
df["order_status"] = df["order_status"].str.strip().str.title()
df["city"] = df["city"].str.strip().str.title()
```

[17]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 101 entries, 0 to 103
Data columns (total 11 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   order_id        101 non-null    str
 1   order_date      97 non-null     datetime64[us]
 2   customer_id     98 non-null     str
 3   product_name    100 non-null    str
 4   category        100 non-null    str
 5   quantity        99 non-null     float64
 6   unit_price      99 non-null     float64
 7   total_amount    99 non-null     float64
 8   payment_method  100 non-null    str
 9   order_status    100 non-null    str
 10  city            100 non-null    str
dtypes: datetime64[us](1), float64(3), str(7)
memory usage: 9.5 KB
```

[18]:
```
df.isnull().sum()
```

[18]:
```
order_id          0
order_date        4
customer_id       3
product_name      1
category          1
quantity          2
unit_price        2
total_amount      2
payment_method    1
order_status      1
```

```
city              1
dtype: int64
```

[19]:
```
df[df["quantity"] < 0]
```

[19]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city
39	ORD1040	2026-02-11	CUST040	Water Bottle	Home & Kitchen	-2.0	12.0	48.0	Upi	Delivered	Ahmedabad

[20]:
```
df[df["unit_price"] < 0]
```

[20]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city
64	ORD1065	2026-03-08	CUST065	Coffee Maker	Home Appliances	2.0	-45.0	90.0	Upi	Delivered	Pune

[21]:
```
df[df["total_amount"] <= 0]
```

[21]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city
42	ORD1043	2026-02-14	CUST043	Gaming Keyboard	Electronics	1.0	89.99	0.0	Upi	Processing	Bengaluru

[22]:
```
def fix_invalid(value):
    if pd.isna(value):
        return np.nan
    if value < 0:
        return np.nan
    return value

df["quantity"] = df["quantity"].apply(fix_invalid)
df["unit_price"] = df["unit_price"].apply(fix_invalid)
df["total_amount"] = df["total_amount"].apply(lambda x: np.nan if pd.notna(x) and x <= 0 else x)

df = df.dropna()
```

[24]:
```
df[(df["quantity"] < 0) | (df["unit_price"] < 0) | (df["total_amount"] <= 0)]
```

[24]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city

[32]:
```
df["calculated_total"] = df["quantity"] * df["unit_price"]
df["total_difference"] = df["total_amount"] - df["calculated_total"]
```

[38]:
```
df["order_date"].isna().sum()
```

[38]:
```
np.int64(0)
```

[39]:
```
df["quantity"].isna().sum()
```

[39]:
```
np.int64(0)
```

[40]:
```
df["unit_price"].isna().sum()
```

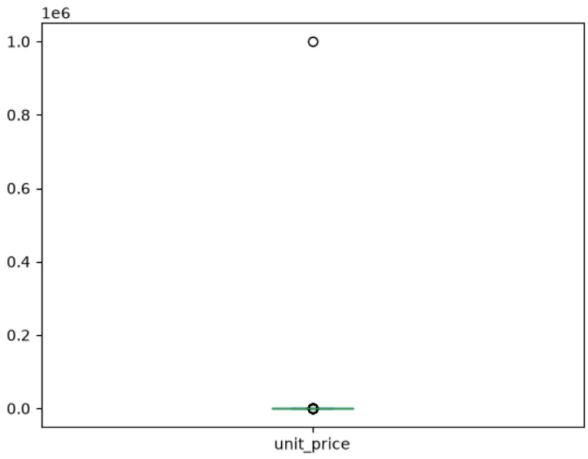
[40]:
```
np.int64(0)
```

[41]:
```
df["total_amount"].isna().sum()
```

[41]:
```
np.int64(0)
```

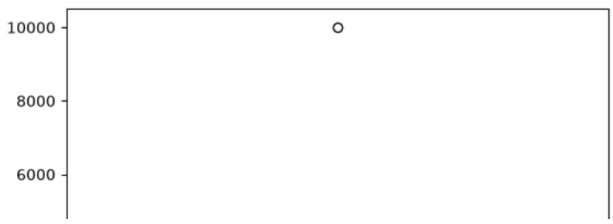
[42]:
```
df["unit_price"].plot(kind="box")
```

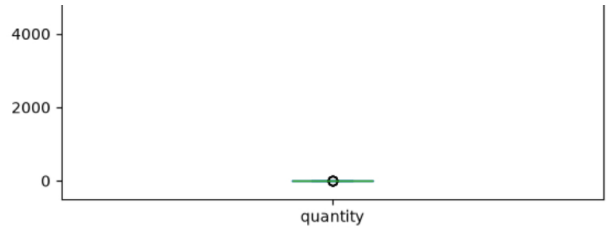
[42]:
```
<Axes: >
```



[43]:
```
df["quantity"].plot(kind="box")
```

[43]:
```
<Axes: >
```





[44]:
```
Q1 = df["unit_price"].quantile(0.25)
Q3 = df["unit_price"].quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
```

[45]:
```
df[(df["unit_price"] < lower_bound) | (df["unit_price"] > upper_bound)]
```

[45]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city	calculated_total	total_difference
0	ORD1001	2026-01-03	CUST001	Laptop	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai	899.99	0.00
15	ORD1016	2026-01-18	CUST016	Laptop	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai	899.99	0.00
33	ORD1034	2026-02-05	CUST034	Smartphone	Electronics	1.0	699.00	699.00	Credit Card	Processing	Hyderabad	699.00	0.00
35	ORD1036	2026-02-07	CUST036	Running Shoes	Fashion	1.0	999999.99	79.99	Debit Card	Shipped	Delhi	999999.99	-999920.00
48	ORD1049	2026-02-20	CUST049	Smartphone	Electronics	1.0	699.00	699.00	Credit Card	Processing	Hyderabad	699.00	0.00
60	ORD1061	2026-03-04	CUST061	Laptop	Electronics	10000.0	899.99	899.99	Credit Card	Shipped	Chennai	8999900.00	-8999000.01
63	ORD1064	2026-03-07	CUST064	Smartphone	Electronics	1.0	699.00	699.00	Credit Card	Processing	Hyderabad	699.00	0.00
75	ORD1076	2026-03-19	CUST076	Wireless Mouse	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai	899.99	0.00
78	ORD1079	2026-03-22	CUST079	Smartphone	Electronics	1.0	699.00	699.00	Credit Card	Processing	Hyderabad	699.00	0.00
93	ORD1094	2026-04-06	CUST094	Smartphone	Electronics	1.0	699.00	699.00	Credit Card	Processing	Hyderabad	699.00	0.00
103	ORD1034	2026-02-05	CUST034	Monitor	Electronics	1.0	699.00	699.00	Credit Card	Processing	Hyderabad	699.00	0.00

[46]:
```
df.at[35,'unit_price']=999
```

[47]:
```
df[(df["unit_price"] < lower_bound) | (df["unit_price"] > upper_bound)]
```

[47]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city	calculated_total	total_difference
0	ORD1001	2026-01-03	CUST001	Laptop	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai	899.99	0.00
15	ORD1016	2026-01-18	CUST016	Laptop	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai	899.99	0.00
33	ORD1034	2026-02-05	CUST034	Smartphone	Electronics	1.0	699.00	699.00	Credit Card	Processing	Hyderabad	699.00	0.00
35	ORD1036	2026-02-07	CUST036	Running Shoes	Fashion	1.0	999.00	79.99	Debit Card	Shipped	Delhi	999999.99	-999920.00
48	ORD1049	2026-02-20	CUST049	Smartphone	Electronics	1.0	699.00	699.00	Credit Card	Processing	Hyderabad	699.00	0.00
60	ORD1061	2026-03-04	CUST061	Laptop	Electronics	10000.0	899.99	899.99	Credit Card	Shipped	Chennai	8999900.00	-8999000.01
63	ORD1064	2026-03-07	CUST064	Smartphone	Electronics	1.0	699.00	699.00	Credit Card	Processing	Hyderabad	699.00	0.00
75	ORD1076	2026-03-19	CUST076	Wireless Mouse	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai	899.99	0.00
78	ORD1079	2026-03-22	CUST079	Smartphone	Electronics	1.0	699.00	699.00	Credit Card	Processing	Hyderabad	699.00	0.00
93	ORD1094	2026-04-06	CUST094	Smartphone	Electronics	1.0	699.00	699.00	Credit Card	Processing	Hyderabad	699.00	0.00
103	ORD1034	2026-02-05	CUST034	Monitor	Electronics	1.0	699.00	699.00	Credit Card	Processing	Hyderabad	699.00	0.00

[48]:
```
Q1 = df["quantity"].quantile(0.25)
Q3 = df["quantity"].quantile(0.75)
IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
```

[49]:
```
df[(df["quantity"] < lower_bound) | (df["quantity"] > upper_bound)]
```

[49]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city	calculated_total	total_difference
26	ORD1027	2026-01-29	CUST027	Notebook Set	Stationery	5.0	4.99	24.95	Cash On Delivery	Delivered	Lucknow	24.95	-3.552714e-15
41	ORD1042	2026-02-13	CUST042	Notebook Set	Stationery	5.0	4.99	24.95	Cash On Delivery	Delivered	Lucknow	24.95	-3.552714e-15
56	ORD1057	2026-02-28	CUST057	Notebook Set	Stationery	5.0	4.99	24.95	Cash On Delivery	Delivered	Lucknow	24.95	-3.552714e-15
60	ORD1061	2026-03-04	CUST061	Laptop	Electronics	10000.0	899.99	899.99	Credit Card	Shipped	Chennai	8999900.00	-8.999000e+06
71	ORD1072	2026-03-15	CUST072	Notebook Set	Stationery	5.0	4.99	24.95	Cash On Delivery	Delivered	Lucknow	24.95	-3.552714e-15
86	ORD1087	2026-03-30	CUST087	Notebook Set	Stationery	5.0	4.99	24.95	Cash On Delivery	Delivered	Lucknow	24.95	-3.552714e-15

[50]:
```
df.at[60,'quantity']=1
```

[51]:
```
df[(df["quantity"] < lower_bound) | (df["quantity"] > upper_bound)]
```

[51]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city	calculated_total	total_difference
26	ORD1027	2026-01-29	CUST027	Notebook Set	Stationery	5.0	4.99	24.95	Cash On Delivery	Delivered	Lucknow	24.95	-3.552714e-15
41	ORD1042	2026-02-13	CUST042	Notebook Set	Stationery	5.0	4.99	24.95	Cash On Delivery	Delivered	Lucknow	24.95	-3.552714e-15
56	ORD1057	2026-02-28	CUST057	Notebook Set	Stationery	5.0	4.99	24.95	Cash On Delivery	Delivered	Lucknow	24.95	-3.552714e-15
71	ORD1072	2026-03-15	CUST072	Notebook Set	Stationery	5.0	4.99	24.95	Cash On Delivery	Delivered	Lucknow	24.95	-3.552714e-15

86	ORD1087	2026-03-30	CUST087	Notebook Set	Stationery	5.0	4.99	24.95	Cash On Delivery	Delivered	Lucknow	24.95	-3.552714e-15

[55]:
```
df.isnull().sum()
```

[55]:
```
order_id           0
order_date         0
customer_id        0
product_name       0
category           0
quantity           0
unit_price         0
total_amount       0
payment_method     0
order_status       0
city               0
calculated_total   0
total_difference   0
dtype: int64
```

[56]:
```
df.duplicated().sum()
```

[56]:
```
np.int64(0)
```

[57]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 80 entries, 0 to 103
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   order_id          80 non-null     str
 1   order_date        80 non-null     datetime64[us]
 2   customer_id       80 non-null     str
 3   product_name      80 non-null     str
 4   category          80 non-null     str
 5   quantity          80 non-null     float64
 6   unit_price        80 non-null     float64
 7   total_amount      80 non-null     float64
 8   payment_method    80 non-null     str
 9   order_status      80 non-null     str
 10  city              80 non-null     str
 11  calculated_total  80 non-null     float64
 12  total_difference  80 non-null     float64
dtypes: datetime64[us](1), float64(5), str(7)
memory usage: 10.8 KB
```

[58]:
```
df.describe()
```

[58]:

	order_date	quantity	unit_price	total_amount	calculated_total	total_difference
count	80	80.000000	80.000000	8.000000e+01	8.000000e+01	8.000000e+01
mean	2026-02-25 02:06:00	1.968750	160.607125	1.251777e+05	1.251555e+05	2.218769e+01
min	2026-01-03 00:00:00	1.000000	4.990000	2.495000e+01	2.495000e+01	-8.999000e+06
25%	2026-02-02 12:00:00	1.000000	22.500000	4.800000e+01	4.800000e+01	0.000000e+00
50%	2026-02-26 12:00:00	2.000000	45.000000	8.997000e+01	8.997000e+01	0.000000e+00
75%	2026-03-20 06:00:00	2.000000	129.990000	1.348675e+02	1.348675e+02	0.000000e+00
max	2026-04-12 00:00:00	5.000000	999.000000	1.000000e+07	8.999900e+06	9.999951e+06
std	NaN	1.212654	264.959799	1.118014e+06	1.010985e+06	1.517747e+06

[59]:
```
df.head()
```

[59]:

	order_id	order_date	customer_id	product_name	category	quantity	unit_price	total_amount	payment_method	order_status	city	calculated_total	total_difference
0	ORD1001	2026-01-03	CUST001	Laptop	Electronics	1.0	899.99	899.99	Credit Card	Shipped	Chennai	899.99	0.0
1	ORD1002	2026-01-04	CUST002	Wireless Mouse	Electronics	2.0	19.99	39.98	Upi	Delivered	Bengaluru	39.98	0.0
2	ORD1003	2026-01-05	CUST003	Office Chair	Furniture	1.0	149.50	149.50	Cash On Delivery	Delivered	Mumbai	149.50	0.0
4	ORD1005	2026-01-07	CUST005	Coffee Maker	Home Appliances	2.0	45.00	90.00	Upi	Delivered	Pune	90.00	0.0
7	ORD1008	2026-01-10	CUST008	Bluetooth Speaker	Electronics	1.0	59.99	59.99	Paypal	Cancelled	Chennai	59.99	0.0

[60]:
```
df.to_csv("ecommerce_cleaned.csv", index=False)
```